Summary in Graph

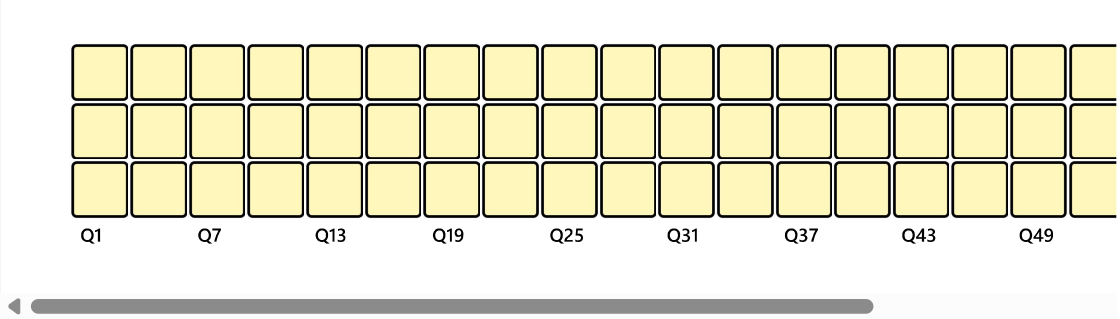
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (36) out of 358 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Two ships, one 200 metres in length and the other 100 metres in length, travel at constant but different speeds. When travelling in opposite directions, it takes 10 seconds for them to completely pass each other. When travelling in the same direction, it takes 25 seconds for them to completely pass each other. The speed of the faster ship, in metres per second, is -

A. 12

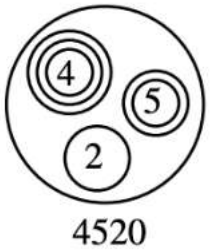
B. 14
C. 18
D. 21

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

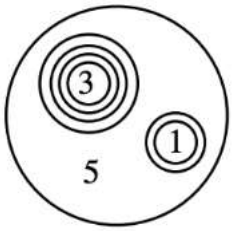
Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

In Circle Land, the numbers 207 and 4520 are shown in the following way:





In Circle Land, what number does the following diagram represent?



A. 30105
B. 30150
C. 3105
D. 3015

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

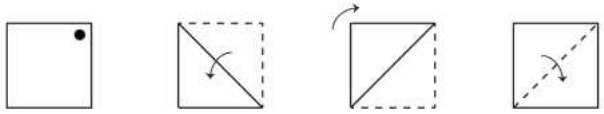
The average (mean) of a list of 10 numbers is 0 . If 72 and -12 are added to the list, the new average will be ?

A. 30
B. 5
C. 0
D. 60

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

A square piece of paper has a dot in its top right corner and is lying on a table. The square is folded along its diagonal, then rotated 90° clockwise about its centre, and then finally unfolded, as shown.



The resulting figure is

A. 

B. 

C. 

D. 

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

The Prime Minister does not approve _____ the views expressed by the official.

A. of
B. at
C. for
D. on

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

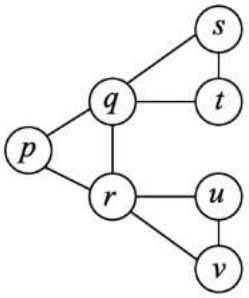
Nasim buys trading cards in packages of 5 cards and in packages of 8 cards. He can purchase exactly 18 cards by buying two 5-packs and one 8 -pack, but he cannot purchase exactly 12 cards with any combination of packages. For how many of the integers $n = 24, 25, 26, 27, 28, 29$ can he buy exactly n cards?

A. 2
B. 3
C. 5
D. 4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 03sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

In the diagram, each of p, q, r, s, t, u, v is to be replaced with 1,2 or 3 so that p, q and r are all different, q, s and t are all different, and r, u and v are all different.
What is the maximum possible value of $s + t + u + v$?



A. 8
B. 9
C. 11
D. 7

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

Each of three cards is labelled with three numbers. Which of the following groups of three cards has the properties that the first and second cards have exactly one number in common, the first and third cards have exactly one number in common, and the second and third cards have exactly one number in common?

A. 135 367 246
B. 147 234 245
C. 234 257 124
D. 147 234 257

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

It takes Pearl 7 days to dig 4 holes. It takes Miguel 3 days to dig 2 holes. If they work together and each continues digging at these same rates, how many holes in total will they dig in 21 days?

A. 35
B. 22
C. 12
D. 26

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

From among the given sentences pick the grammatically correct one:

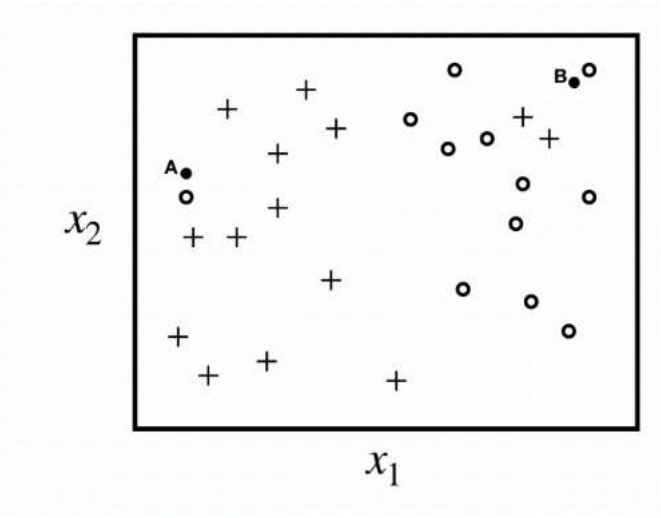
A. Hugging his mother he opened the gift box.
B. His performance in the match was superior than his teammates.
C. I saw them playing while driving down the street.
D. Laughing out loud Teja tried to express his happiness.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a binary classification dataset with two-dimensional input vectors ($\mathrm{x1}, \mathrm{x2}$) and labels + and °. Two new points, $\mathbf{A}$ and $\mathbf{B}$, are introduced for classification. Assume the K-Nearest Neighbors (K-NN) algorithm uses Euclidean distance to make predictions.



What are the predicted labels for points $\mathbf{A}$ and $\mathbf{B}$ under the 1-NN classifier?

A. 1-NN predicts + for A and + for B .
B. 1-NN predicts + for A and ^{O} for B .
C. 1-NN predicts ° for A and + for B .
D. 1-NN predicts ^{O} for A and ^{O} for B .

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

We trained a three-way softmax regression model and obtained the following weights for the classes:
- $w_a = (1, 1, 0)$,
- $w_b = (-1, 1, 1)$,
- $w_c = (2, 1, 2)$.

The softmax regression model computes the probabilities for each class $i \in \{a, b, c\}$ using the formula:

$$P(y = i \mid x) = \frac{e^{w_i^\top x}}{\sum_{j \in \{a,b,c\}} e^{w_j^\top x}}$$

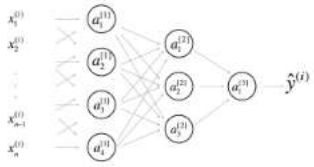
What label would be assigned to the point $x = (0, 1, 1)$?

A. Class A
B. Class B
C. Class C
D. Cannot be determined

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You design the following 2-layer fully connected neural network.



All activations are sigmoids and your optimizer is stochastic gradient descent. You initialize all the weights and biases to zero and forward propagate an input $x \in \mathbb{R}^{n \times 1}$ in the network. What is the output $\hat{y}$?

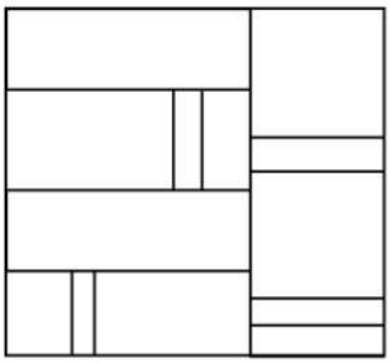
A. -1
B. 0
C. 0.5
D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

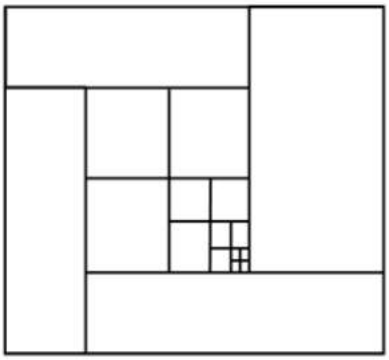
Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following is NOT possible to get partition space using decision tree ?

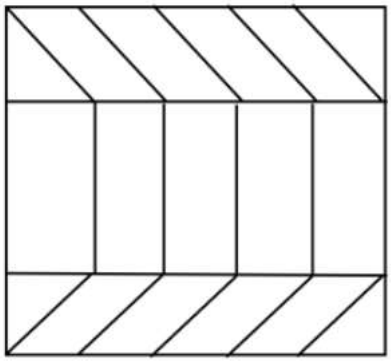
A.



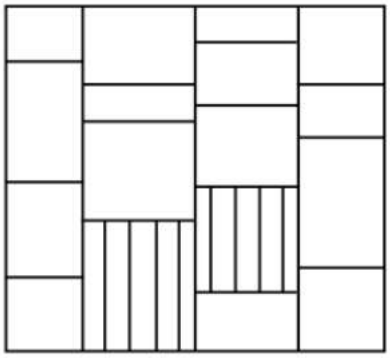
B.



C.



D.



Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Others

Suppose 10 balls are thrown into 10 bins, where each ball is equally likely to land in any one of the 10 bins. The bins are numbered 1 to 10. Define: - X_1 : Number of balls that land in bin 1. - X_2 : Number of balls that land in bin 2. - Similarly, $X_3, X_4, \ldots, X_{10}$. Let $W = X_1 + X_2 + \ldots + X_{10}$. What is the variance of W ?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let $X_1, X_2, \ldots, X_n \sim \text{iid Bernoulli}(p)$, and define $S_n = \sum_{i=1}^{n} X_i$. Which of the following statements correctly describes the type of random variable S_n and its support?

A. $S_n \sim \text{Binomial}(n, p)$, and its support is $\{0, 1, \ldots, n\}$.
B. $S_n \sim \text{Poisson}(\lambda = np)$, and its support is $\{0, 1, 2, \ldots\}$.
C. $S_n \sim \text{Uniform}(0, n)$, and its support is $\{0, 1, \ldots, n\}$.
D. $S_n \sim \text{Exponential}(\lambda = 1/p)$, and its support is $[0, \infty)$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

An experiment has a sample space $\Omega = \{o_1, o_2, o_3, o_4, o_5\}$, and the probability rule is defined as:

$$P(o_1) = \frac{1}{4}, \quad P(o_2) = \frac{1}{4}, \quad P(o_3) = \frac{1}{4}, \quad P(o_4) = \frac{1}{8}, \quad P(o_5) = \frac{1}{8}.$$

The random variable X is defined as:

$$X(o_1) = 1, \quad X(o_2) = 1, \quad X(o_3) = 2, \quad X(o_4) = 2, \quad X(o_5) = 2.$$

What is the value of $P(X = 2)$?

A. $1/4$
B. $1/2$
C. $3/8$
D. $5/8$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let X and Y be independent random variables. Then

A. $\mathrm{Var}[X - Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$
B. $\mathrm{Var}[X - Y] = \mathrm{Var}[X] - \mathrm{Var}[Y]$
C. $\mathrm{Var}[X - Y]^2 = \mathrm{Var}[X]^2 + \mathrm{Var}[Y]^2$
D. $\mathrm{Var}[X - Y]^2 = \mathrm{Var}[X]^2 - \mathrm{Var}[Y]^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Given that $X \sim \mathrm{Normal}(\mu = 3, \sigma = 2)$, the probability $P(-1 \leq X \leq 2)$ can be converted to a standard normal distribution Z. Which of the following correctly represents this conversion?

A. $P(-1 \leq X \leq 2) = P\left(-2 \leq Z \leq -\frac{1}{2}\right)$
B. $P(-1 \leq X \leq 2) = P(-1 \leq Z \leq 2)$
C. $P(-1 \leq X \leq 2) = P(1 \leq Z \leq 2.5)$
D. $P(-1 \leq X \leq 2) = P(1 \leq Z \leq 2)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose A is an $m \times n$ matrix, and the row space of A is n-dimensional, then its nullspace is _____ dimensional.

A. 0
B. m
C. n
D. $\mathrm{n} - 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Let A be an $n \times n$ matrix. Which of the following properties would necessarily imply that A is singular?

A. The columns of A are linearly dependent.
B. A has a singular value that is 0 .

C. $Az = 0$, for some z not equal to 0 .
D. The determinant of A is nonzero.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Others

What is the rank of $\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{pmatrix}$?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Compute $\lim_{x\to\infty} \left(1 - \frac{3}{x}\right)^x$.

A. e^3.
B. 0.
C. e^{-3}.
D. 1.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

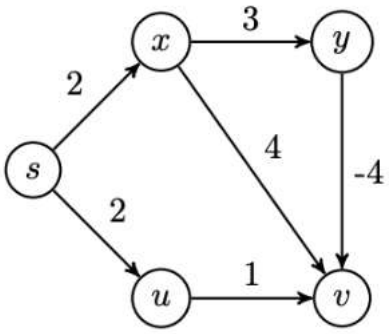
Given a binary search tree T which has n nodes and height h, what is the time required to find an item is T ? Fill in one circle.

A. $O(n)$
B. $O(h)$
C. $O(\log n)$
D. $O(\log h)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Consider the directed graph



In this graph, if we run Dijkstra's algorithm what is the distances that we find from s to v ?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider an array of size n . You need to **implement two stacks** using a single array. The first stack inserts elements starting from position 0, moving to the right, while the second stack inserts elements starting from position $n-1$, moving to the left. You need to push and pop elements from both stacks in constant time and detect overflow when the combined size of both stacks exceeds n . Which of the following is correct?

A. Overflow occurs when T1 + T2 == n, where T1 is the top of the first stack and T2 is the top of the second stack.
B. Overflow occurs when T2 - T1 == 1, where T1 is the top of the first stack and T2 is the top of the second stack.
C. Both stacks grow towards the center of the array, and overflow occurs when T1 == T2.
D. The two stacks cannot be implemented in constant time in a single array.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Others

Consider the sequence [10, 20, 30, 40, 50, 60, 70, 80, 90]. If you're searching for the number 35 using binary search, how many comparisons will you need before determining that 35 is not in the list?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the given function last_greater( ) :

```
    def last_greater(lst, num):
        last_index = None
        for i in range(0, len(lst)):
            if lst[i] >= num:
5.              last_index = i
        return last_index
```

What will be the output of the following call to the function last_greater([10, 20, 30, 11, 13], 12) ?

A. 2

B. 3
C. 4
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following python dict and python function:

```
my_dict = {'a': 4, 'b': 5, 'd': 7}
def manipulate_dict(input_dict):
    input_dict['c'] = 6
    for key in input_dict:
5.      input_dict[key] += 2
    input_dict['d'] = 0
    return input_dict
```

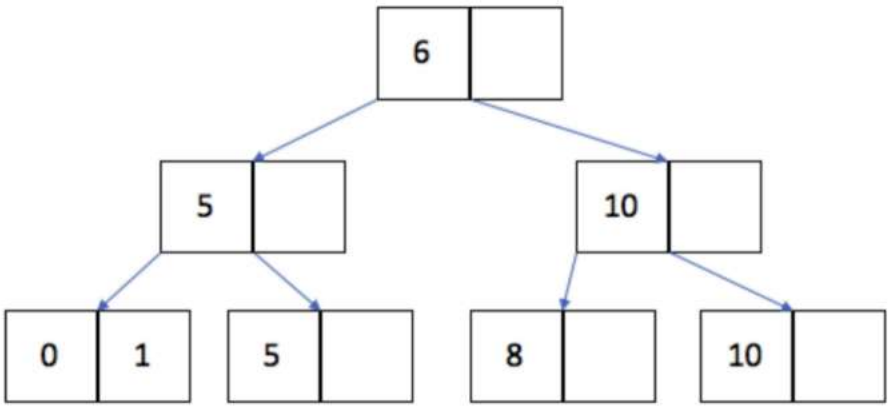
What will the dictionary my_dict look like after executing the function manipulate_dict(my_dict)?

A. {'a': 6, 'b': 7, 'd': 9, 'c': 6}
B. {'a': 6, 'b': 7, 'c': 6}
C. {'a': 6, 'b': 7, 'd': 0, 'c': 8}
D. {'a': 4, 'b': 5, 'c': 6}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following B+-tree with order $d = 1$, and leaves that hold 2 entries.



What is the maximum number of inserts we can do without changing the height of the tree?

Your Answer: | Correct Answer: 13 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following table named account:

witness	suspect

Bob	Peter
Peter	Bob
Queen	Bob

What will the following SQL query return?

```
SELECT DISTINCT witness
FROM account
WHERE witness NOT IN (SELECT suspect FROM account);
```

A. Queen
B. Bob and Queen
C. Peter and Queen
D. None

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Suppose we have a boolean variable X. To complete describe the distribution $P(X)$, we need to specify one value: $\mathrm{P}(\mathrm{X}=0)$ (since $\mathrm{P}(\mathrm{X}=1)$ is simply 1- $\mathrm{P}(\mathrm{X}=0)$) Thus, we say, this distribution can be characterized with

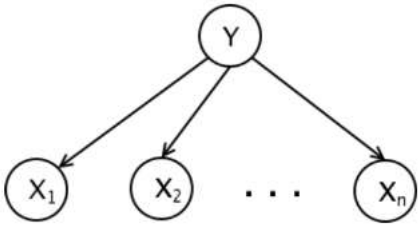


Figure 1: Bayesian network for Problem

one parameter. Now, consider $\mathrm{N}+1$ binary random variables $X_1 \ldots X_N, Y$ that factorize according to Fig. 1 Now, suppose you were to utilize the fact the joint distribution factorizes accoording to the Bayes Network. How many parameters will you need to completely describe the distribution if you use the Bayesian Network representation? In other words, how many parameters will you need to fully specify the values of all the conditional probability tables in this Bayesian Network.

A. $2^{N+1}-1$
B. 2N
C. $2\mathrm{N}+1$
D. $\mathrm{N}+1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following statements is/are TRUE?

A. Breadth-first search and iterative-deepening search always find the same solution
B. Breadth-first search is a special case of uniform cost search
C. A heuristic that always evaluates to $h(s)=1$ for non-goal search nodes s is always admissible
D. Greedy graph search is guaranteed to return an optimal solution.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Let H1 and H2 both be admissible heuristics. Which of the following statements is FALSE with respect to a heuristic function?

A. $\max(\mathrm{H1}, \mathrm{H2})$ is necessarily admissible
B. $\min(\mathrm{H1}, \mathrm{H2})$ is necessarily admissible
C. $(\mathrm{H1} + \mathrm{H2})/2$ is necessarily admissible
D. $\max(\mathrm{H1}, \mathrm{H2})$ is necessarily consistent

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

The function $((p \vee (r \vee q)) \wedge \sim (\sim q \wedge \sim r)$ is equal to the function:

A. $q \vee r$
B. $((p \vee r) \vee q)) \wedge (p \vee r)$
C. $(p \wedge q) \vee (p \wedge r)$
D. $(p \vee q) \wedge \sim (p \vee r)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A logistic regression model with parameters $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$ is trained on a dataset of points $x \in \mathbb{R}^d$ with binary labels $y \in \{-1, 1\}$. The probability of $y = 1$ is given by:

$$\Pr(y = 1 \mid x) = \frac{1}{1 + \exp(-w \cdot x)}$$

Find the set of points x that satisfy $\Pr(y = 1 \mid x) = 3/4$.

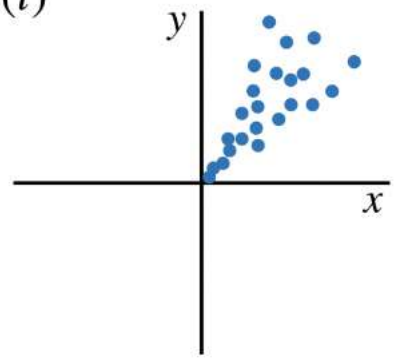
A. $w \cdot x + \log(3/4) = 0$
B. $w \cdot x + \log(4/3) = 0$
C. $w \cdot x + \log(1/3) = 0$
D. $w \cdot x = \log(3/4)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

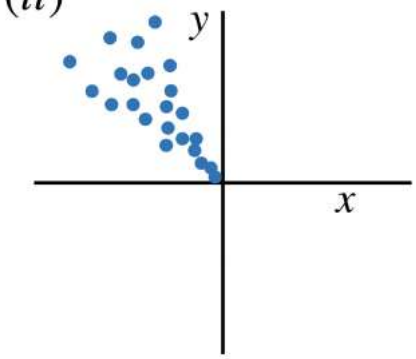
Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the two regression training datasets depicted in plots (i) and (ii). Plot (ii) is exactly the mirror image of plot (i) around the y-axis.

(i)



(ii)



Consider two loss functions:

Mean Squared Error (MSE)	Mean Absolute Error (MAE)
$MSE(w)$ = $\frac{1}{n}\sum_{i=1}^{n}(y^{(i)} - w.x^{(i)})^2$	$MAE(w)$ = $\frac{1}{n}\sum_{i=1}^{n}\|(y^{(i)} - w.x^{(i)})\|$

We find four optimal weights (w):
- Two for plot (i): one for MSE and one for MAE,
- Two for plot (ii): one for MSE and one for MAE.
Will plots (i) and (ii) result in the same training loss after finding the best weight w ?

A. Plots (i) and (ii) have the same training MSE. Plots (i) and (ii) have the same training MAE.
B. Plots (i) and (ii) have the same training MSE. Plots (i) and (ii) have different training MAE.
C. Plots (i) and (ii) have different training MSE. Plots (i) and (ii) have the same training MAE.
D. Plots (i) and (ii) have different training MSE. Plots (i) and (ii) have different training MAE.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the following loss function on vectors $w \in \mathbb{R}^4$:

$$L(w) = w_1^2 + 2w_2^2 + w_3^2 - 2w_3w_4 + w_4^2 + 2w_1 - 4w_2 + 4$$

Suppose we use gradient descent to minimize this function, starting with the current estimate $w = (0, 0, 0, 0)$. If the step size is η, what will be the next estimate for w ?

A. $(-2\eta, 4\eta, \eta, -\eta)$
B. $(-2\eta, 4\eta, 0, 0)$
C. $(2\eta, -4\eta, 0, 0)$

D. $(-2\eta, 4\eta, -\eta, \eta)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider a set of linearly separable training examples. The perceptron algorithm is trained twice with different initial weights. Both training procedures traverse the data points in the same order and are run until convergence.
Which of the following statements is correct?

A. Both classifiers will have the same performance on the training set but may have different performance on the test set.
B. Both classifiers will have the same performance on both the training and test sets.
C. The classifiers may have different performance on the training set but the same performance on the test set.
D. The classifiers may have different performance on both the training and test sets.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

You are tasked with learning $P(\mathrm{Y} \mid \mathrm{X1}, \mathrm{X2}, \mathrm{X3})$, where $\mathrm{Y}, \mathrm{X1}, \mathrm{X2}$, and $\mathrm{X3}$ are boolean-valued random variables. Consider using Naive Bayes and Logistic Regression as possible approaches.
How many parameters must be estimated for each model?

A. Naive Bayes: 7 parameters, Logistic Regression: 4 parameters
B. Naive Bayes: 6 parameters, Logistic Regression: 3 parameters
C. Naive Bayes: 8 parameters, Logistic Regression: 5 parameters
D. Naive Bayes: 7 parameters,Logistic Regression: 3 parameters

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Given a pairwise distance matrix D, where D_{ij} is the distance between points x_i and x_j, and two clusters $G = \{x_i\}$ and $H = \{x_j\}$, what is the formula to compute the distance $d(G, H)$ between the two clusters using average linkage?

A. $d(G, H) = \min_{x_i \in G, x_j \in H} D_{ij}$
B. $d(G, H) = \max_{x_i \in G, x_j \in H} D_{ij}$
C. $d(G, H) = \frac{1}{|G||H|} \sum_{x_i \in G} \sum_{x_j \in H} D_{ij}$
D. $d(G, H) = \frac{\sum_{x_i \in G} \sum_{x_j \in H} D_{ij}}{|G|+|H|}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let X and Y be two jointly continuous random variables with the joint PDF:

$$f_{X,Y}(x, y) = \begin{cases} x + y & \text{for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{E}\left[XY^2\right]$.

A. $\frac{1}{8}$
B. $\frac{17}{72}$
C. $\frac{1}{6}$
D. $\frac{19}{72}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let the joint PDF of X and Y be:

$$f_{X,Y}(x,y) = \begin{cases} x + \frac{3}{2}y^2 & \text{for } 0 \le x \le 1, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{E}[X]$.

A. $\frac{1}{2}$
B. $\frac{7}{12}$
C. $\frac{5}{6}$
D. $\frac{3}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let X and Y be two jointly continuous random variables with the joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} x + cy^2 & \text{for } 0 \le x \le 1 \text{ and } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{P}\left(0 \le X \le \frac{1}{2}, 0 \le Y \le \frac{1}{2}\right)$.

A. $\frac{1}{16}$
B. $\frac{1}{8}$
C. $\frac{3}{32}$
D. $\frac{1}{4}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given the following joint probability distribution of X and Y :

	$X=1$	$X=2$
$Y=0$	0.30	0.20
$Y=1$	0.25	0.25

Compute the expected values $\mathbb{E}[Y \mid X=1]$ and $\mathbb{E}[Y \mid X=2]$.

A. $\mathbb{E}[Y \mid X=1]=6/11, \mathbb{E}[Y \mid X=2]=4/9$
B. $\mathbb{E}[Y \mid X=1]=5/9, \mathbb{E}[Y \mid X=2]=5/11$
C. $\mathbb{E}[Y \mid X=1]=4/11, \mathbb{E}[Y \mid X=2]=6/9$
D. $\mathbb{E}[Y \mid X=1]=5/11, \mathbb{E}[Y \mid X=2]=5/9$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Numerical Type | Award: 2 | Penalty: 0 | Others

If X_1, X_2, X_3, and X_4 are pairwise uncorrelated random variables, each having mean 0 and variance 1 , what is the covariance between X_1+X_2 and X_2+X_3 ?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

If P is the projection matrix onto the null space of A, then $P\mathbf{y}-\mathbf{y}$, for any $\mathbf{y}$, is in the:

A. Row space of A.
B. Column space of A.
C. Null space of A.
D. Left null space of A.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let $T: R^3 \rightarrow R^2$ be a linear transformation such that:

$$T\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}2\\6\end{bmatrix}, \quad T\begin{bmatrix}1\\0\\0\end{bmatrix}=\begin{bmatrix}1\\2\end{bmatrix}, \quad T\begin{bmatrix}2\\1\\3\end{bmatrix}=\begin{bmatrix}7\\20\end{bmatrix}$$

Find the matrix A of the transformation T.

A. $A=\begin{bmatrix}1 & -1 & 2\\2 & -2 & 6\end{bmatrix}$
B. $A=\begin{bmatrix}2 & -1 & 2\\6 & -2 & 6\end{bmatrix}$
C. $A=\begin{bmatrix}1 & -2 & 2\\2 & -1 & 6\end{bmatrix}$
D. $A=\begin{bmatrix}1 & -1 & 1\\2 & -2 & 3\end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Which of the following statements about the Singular Value Decomposition (SVD) are true?

A. Every real matrix has an SVD.
B. If a matrix Q is orthogonal, then its singular values are all 1.
C. A matrix with rank r will have exactly r singular values that are greater than 0 .

D. The SVD of a square matrix is unique.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Which of the following statements about the Least Squares (LS) problem

$$\min_{\mathbf{x}} \|\mathbf{b} - A\mathbf{x}\|_2^2$$

are true?

A. The solution of the LS problem satisfies $A^T A\mathbf{x} = A^T\mathbf{b}$.
B. The solution of the LS problem always exists.
C. The solution of the LS problem is always unique.
D. If A is invertible, then the solution to the LS problem is $\mathbf{x} = A^{-1}\mathbf{b}$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let

$$A = \begin{pmatrix} 2 & t & 0 \\ t & 2 & t \\ 0 & t & 2 \end{pmatrix}$$

Find all values of t for which the matrix A has 3 positive eigenvalues.

Hint: You may use these two equivalent properties of a positive definite matrix, which imply each other:
1. All eigenvalues of the matrix are positive.
2. All leading principal minors (upper-left determinants) are positive.

A. $-2 < t < 2$
B. $-\sqrt{2} < t < \sqrt{2}$
C. $t > \sqrt{2}$
D. $t < -\sqrt{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider the following properties related to optimization and matrix calculus. Identify which of these statements are TRUE based on theoretical foundations of optimization, gradient methods, and matrix properties.

A. A critical point $\mathbf{x}^*$ of f is a minimizer of f if the Hessian matrix $H_f(\mathbf{x}^*)$ is negative definite.
B. A necessary condition for f to have a minimum at $\mathbf{x}^*$ is that $\nabla f(\mathbf{x}^*) = \mathbf{0}$.
C. The Steepest Descent method for finding a minimum of a function $f(\mathbf{x})$ uses $-\nabla f(\mathbf{x}_k)$ as the search direction for a 1D line search.
D. The Hessian matrix $H_f(\mathbf{x})$ being positive semidefinite is sufficient for $\mathbf{x}$ to be a local minimum of f.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

On which of the following intervals is function $f(x) = x^3 + x^2 - x$ both decreasing and concave up?

A. $\left(-\frac{1}{3}, \frac{1}{3}\right)$.
B. $\left(-1, -\frac{1}{3}\right)$.
C. $\left(\frac{1}{3}, \infty\right)$.
D. $(-\infty, -1)$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Find the time complexity of the following code fragment in terms of n using Θ-notation:

```
for i in range(0, n):
    j = i
    while j > 0:
        j = j // 2
```

A. $\Theta\left(n^2\right)$
B. $\Theta(n)$
C. $\Theta(n \log \log n)$
D. $\Theta(n \log n)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A k-sorted array is a nearly sorted array in which no element is more than k locations away from its final position in the sorted array. Thus, a 0 -sorted array is completely sorted and every array of size n is n-sorted. Suppose that A is a k-sorted array of size n. If insertion sort is used to sort A, what is the order of growth of the number of comparisons performed by the sorting algorithm in the worst case?

A. $\Theta(k)$
B. $\Theta(kn)$
C. $\Theta\left(k^2 n\right)$
D. $\Theta\left(n \log_k n\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $d[]$ be the discovery time and $f[]$ be the finish time for each vertex in a depth-first search (DFS) of a directed graph G. For both forward and tree edges $e = (u, v)$, the condition $d[u] < d[v] < f[v] < f[u]$ holds. Using only d[] and f[] , how can you determine if $e = (u, v)$ is a forward or tree edge?

A. If there exists a vertex z such that $d[u] < d[z] < d[v] < f[v] < f[z] < f[u]$, then $e = (u, v)$ is a forward edge; otherwise, it is a tree edge.
B. If $d[u] < d[v] < d[z] < f[v] < f[z] < f[u]$, then $e = (u, v)$ is a forward edge; otherwise, it is a tree edge.
C. If $d[u] < d[v]$ but $f[u] < f[v]$, then $e = (u, v)$ is a forward edge.
D. If $f[v] = d[u] + 1$, then $e = (u, v)$ is a tree edge; otherwise, it is a forward edge.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

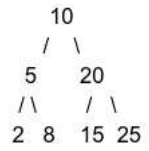
You are given two sorted lists L_1 and L_2 with lengths N_1 and N_2, respectively. Design two algorithms to output the sorted list $L_1 \cap L_2$ (intersection of the two lists):
1. Algorithm 1: Runs in $O(N_1 + N_2)$ time.
2. Algorithm 2: Runs in $O(\min(N_1, N_2) \log(\max(N_1, N_2)))$ time.
Which of the two algorithms is faster?

A. The first algorithm is always faster.
B. The second algorithm is always faster.
C. The first algorithm is faster when $N_1 = \Theta(N_2)$, and the second algorithm is faster when $N_1 = o(N_2)$.
D. Both algorithms have the same time complexity.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a perfect binary tree represented through an array. For example, the following binary



can be represented in an array as:
tree = [10, 5, 20, 2, 8, 15, 25]
Now, consider the function count_left_first(tree) defined as follows:

```
    def count_left_first(tree):
        total = 0
        n = len(tree)
        for i in range(1, n // 2 + 1)
5.          left = 2 * i - 1  # Left child index (convert to 0-based indexing)
            right = 2 * i     # right child index
            if left < n and right < n:  # Check if both children exist
                if tree[left] < tree[i - 1] and tree[left] < tree[right]:
                    total += 1
10.     return total
```

What does this method count_left_first(tree) count?

A. The method counts all nodes where the left child's value is smaller than the parent node's value.

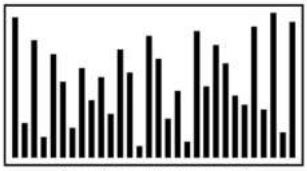
B. The method counts all nodes where the left child's value is smaller than both the parent's value and the right child's value.
C. The method counts all nodes where both children exist, and the right child's value is smaller than the parent's value.
D. The method counts all nodes where the left child's value is greater than the parent node's value.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

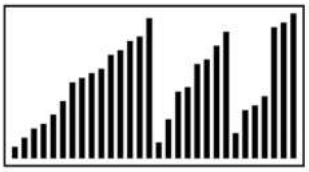
Suppose we want to sort a collection of sticks according to height.
We have observed the intermediate result shown in the image, where sticks are partially sorted during the process.

From:



Starting arrangement

To:



Intermediate Result

Based on this observation, which sorting algorithm are we most likely applying?

A. Bubble Sort
B. Quick Sort
C. Selection Sort
D. Merge Sort

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following recursive Python function:

```
def mystery_function(S, start, stop):
    if start < stop-1:
        S[start], S[stop-1] = S[stop-1], S[start]
        mystery_function(S, start+1, stop-1)
```

What will be the output of the function on the following S when called as mystery_function(S, 1, 6)?
$S = [1, 2, 3, 4, 5, 6, 7]$.

A. $[1, 2, 6, 3, 5, 4, 7]$
B. $[1, 5, 4, 3, 2, 6, 7]$
C. $[1, 6, 5, 4, 3, 2, 7]$
D. $[1, 2, 3, 4, 5, 6, 7]$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider two relations $R(A, B)$ and $S(C, D)$.
Which of the following statements is/are CORRECT

A. The size of $R \bowtie_{A=C} S$ is always larger than or equal to the size of $R \bowtie_{\text{A=C and B=D}} S$.
B. The size of $R \bowtie_{\text{A=C and B}\neq\text{D}} S$ is always larger than or equal to the size of $R \bowtie_{\text{A=C and B=D}}$ S.
C. The size of $R \bowtie_{A=C} S$ is always equal to the size of $R \bowtie_{\text{A=C and B=D}} S$.
D. The size of $R \bowtie_{\text{A=C and B=D}} S$ is always larger than or equal to the size of $R \bowtie_{\text{A=C and B}\neq D}$ S.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider the relation R with the following functional dependencies (FDs), collectively denoted as F :

$$F = \{T \to UV, \quad TV \to WX, \quad UX \to Y\}$$

Determine which of the following statements is/are TRUE.
Select all that apply:

A. $V \to X$ is in the closure F^+.
B. $T \to Y$ is in the closure F^+.
C. TV is a superkey.
D. T is a candidate key.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Numerical Type | Award: 2 | Penalty: 0 | Others

You are given sentences in a language that you know nothing about, where the word order has been lost. Given the words, you are requested to put them back in a sequence, that has a sensible meaning. For example, in English language, the words {fun, mock, are, tests, GOclasses} can be put together in this valid ordering: "GOclasses mock tests are fun". As stated, the language used is completely unknown to you. Luckily, you can use an oracle that assigns a score to every sequence you introduce, according to how ridiculous it is (for example, in English, the sequence "mock tests" is considered less ridiculous than "tests mock"). Given a set of 5 words, what is the problem search space size? (you may assume that all words are unique.)

Your Answer: | Correct Answer: 120 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Given the Bayes net shown in the Figure below; A, B, C, D, and E are all Boolean variables. $\mathrm{P}(\mathrm{A} = \text{"}+\text{"})$ is simply denoted as $\mathrm{P}(\mathrm{A})$.

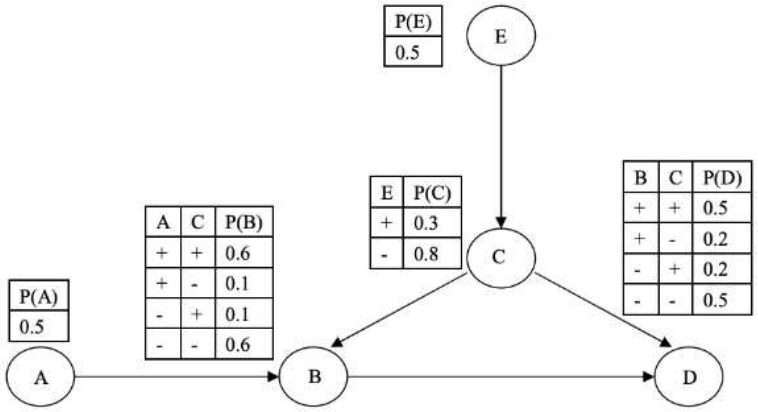


Figure 2: Bayes Net

Note: In the following, we'll use the notations: $A \perp B$ means A is independent of B; $A \perp B \mid C$ means A is conditionally independent of B given C .

Which of the following independence assumptions are correct?

A. $B \perp E \mid C$
B. $A \perp D$
C. $A \perp D \mid B$
D. $A \perp D \mid B, C$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Which of the following formulas is a formalization of the sentence: "No dogs are intelligent"

1. $\forall \mathrm{x}(\mathrm{dog}(\mathrm{x}) \wedge \neg \mathrm{Intelligent}(\mathrm{x}))$
2. $\forall \mathrm{x}(\mathrm{Intelligent}(\mathrm{x}) \rightarrow \neg\, \mathrm{dog}(\mathrm{x}))$
3. $\forall x(\mathrm{dog}(\mathrm{x}) \rightarrow \neg \mathrm{Intelligent}(\mathrm{x}))$
4. $\neg \exists x(\mathrm{dog}(\mathrm{x}) \wedge \mathrm{Intelligent}(\mathrm{x}))$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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